

Developing dynamic capabilities in electronic marketplaces: A cross-case study

Hope Koch

Information Systems Department, Baylor University, One Bear Place #98005, Waco, TX 76798-8005, United States

ARTICLE INFO

Article history:

Received 5 May 2009

Accepted 1 February 2010

Keywords:

Electronic markets

Dynamic capabilities

Industry competition

Network

Capability development

Boundary spanner

ABSTRACT

Business-to-business (B2B) electronic marketplaces (EMPs) have increased the efficiency and economy of business purchasing by bringing together a critical mass of organizations that buy and sell goods and services. While many studies have investigated mature EMPs and identified factors leading to their success and failure, few studies have investigated the capabilities necessary to *develop* successful EMPs. This research shares findings from in-depth case studies of two EMPs conducted over a 3-year period. Using the dynamic capabilities framework as a theoretical lens, this paper: (1) identifies the capabilities necessary to develop EMPs that generate and sustain participant contributions, and (2) discusses how to develop these capabilities. This study finds that the cultivation of a “trial-and-error” culture along with sales managers’ activities played key roles in developing outside-in and spanning capabilities. Taken together these capabilities helped the successful EMP develop entrepreneurial alertness and customer agility, two capabilities that were not developed in the failed EMP. These findings extend dynamic capabilities theory and may help practitioners better develop two-sided networks, such as EMPs, that require a critical mass of buyers and sellers.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

Business-to-business (B2B) electronic marketplaces (EMPs) are Internet-based intermediary hubs that bring together a critical mass of organizations that buy and sell products and services (Malone et al., 1987; Matook and Vessey, 2008). These EMPs offer many advantages over traditional hierarchical modes of coordination, including reducing search costs, increasing competition, lowering product costs, and reducing coordination costs (Bakos, 1997; Granados et al., 2007).

Many researchers have studied the success and failure of EMPs (Allen et al., 2000; Premkumar et al., 2004; Howard et al., 2006; Soh et al., 2006; White et al., 2007). These studies conclude that EMPs that succeed attract a critical mass of active members, while EMPs that fail do not. Attracting a critical mass of active members requires EMPs to offer both sides of the network (that is, both buyers and sellers) value. Unfortunately, many EMPs have failed (Day and Fein, 2003) because parties in a trading relationship often believe that EMPs may leave them worse off (Mithas et al., 2008). Common concerns are that EMPs may harm existing business relationships (Ganesh and Madanmohan, 2004; Howard et al., 2006; Mithas et al., 2008), increase transaction risk (Barrett and Walsham, 1999; Allen et al., 2000; Howard et al., 2006), and/or increase price transparency (Ganesh and Madanmohan, 2004; Soh et al., 2006).

Many organizations that have overcome these challenges and developed valuable offerings have the advantage of migrating existing auctions (Lee and Clark, 1996; Kambil and van Heck, 1998), markets (Choudhury et al., 1998; Barrett and Walsham, 1999; Grewal et al., 2001), or buyer–seller networks (Hess and Kemerer, 1994; Premkumar et al., 2004; Soh et al.,

E-mail address: Hope_Koch@baylor.edu

2006) to EMPs. However, other organizations develop EMPs without the benefits of a pre-existing membership base and already-developed offerings or processes. With the exception of Rangan (1999) and Ganesh and Madanmohan (2004), few studies have investigated the capabilities that these organizations focus on developing in order to create EMPs that add value and therefore attract a critical mass of active members. Understanding the process of developing EMPs that become value-adding intermediaries can provide much needed insight into their success and failure.

This study seeks to address these gaps by reporting on longitudinal case studies of two B2B EMPs, each of which evolved its business model over the course of many years. Using within- and cross-case analyses, this study answers the following research questions:

- What types of capabilities differentiate organizations that create B2B EMPs that are able to generate and sustain participant contributions from those that do not?
- How do executives create these capabilities within an EMP's organizational structure?

This paper is organized as follows. Drawing primarily on the dynamic capabilities framework (Teece et al., 1997; Sambamurthy et al., 2003), the paper outlines the theoretical basis for the argument that the success of EMPs can be explained by the capabilities that the executives focus on developing. The paper then explains the research method, outlines each EMP's evolution, and contrasts the development efforts of each EMP's founding executives. Based on this analysis, the discussion identifies: (1) the capabilities that differentiated the EMP that succeeded from the EMP that failed, and (2) how the executives of the successful EMP developed these capabilities within their organization.

2. Theoretical background: dynamic capabilities

This research employs the dynamic capabilities framework (Sambamurthy et al., 2003) as a theoretical lens to answer the research questions. As capability-building mechanisms, dynamic capabilities are a “firm's processes that use resources – specifically the processes to integrate, reconfigure, gain, and release resources – to match and even create market change” (Eisenhardt and Martin, 2000, p. 1107). In the information systems (IS) field, significant dynamic capabilities include digitized process reach, customer agility, and entrepreneurial alertness (Sambamurthy et al., 2003). Digitized process reach is an organization's ability to deploy integrated and connected information technology (IT) enabled processes like B2B EMPs. This ability is influenced by an organization's customer agility and entrepreneurial alertness. Customer agility, which is created from a customer-centric culture that fosters trust and motivates customer involvement, is the ability to co-opt customers as partners in exploring and exploiting opportunities. Organizations create entrepreneurial alertness, which is the ability to explore marketplaces and identify opportunities for action, by probing and trial-and-error experimentation.

As higher-order capabilities, dynamic capabilities emerge over time from bundles of capabilities (Sambamurthy et al., 2003). Capabilities bring assets like IT infrastructure together and enable an organization to deploy them advantageously (Day, 1994). An organization's capabilities can be sorted into inside-out, outside-in, and spanning capabilities (Day, 1994; Wade and Hulland, 2004). Inside-out capabilities are deployed from within the organization, tend to be internally-focused, and include abilities like technology development and employee recruitment. Outside-in capabilities are oriented externally and include external relationship management and market responsiveness. Spanning capabilities integrate inside-out and outside-in capabilities. Strategy development and new product/service development are activities that must be informed by both external (outside-in) and internal (inside-out) analyses (Day, 1994). Executives and boundary spanners (e.g., sales agents and purchasing managers) develop outside-in and spanning capabilities by assuming cross-organizational roles (Montealegre, 2002; Sambamurthy et al., 2003; Levina and Vaast, 2005) through which they develop embedded relationships with outside organizations.

Dynamic capabilities are typically used to study how a single organization secures an advantage over its competitors (Teece et al., 1997). However, in cases involving B2B EMPs that serve an entire industry, cooperation – the process of working together to achieve a common goal (Teece, 1992) – plays a key role in whether, and how, capabilities can be developed. Since developing successful EMPs requires funding, business process knowledge, and a critical mass of industry organizations that use the EMPs on a regular basis (Soh et al., 2006), organizations creating EMPs require this type of cooperation throughout the EMPs' life cycle. Towards this end, this paper analyzes the process by which two industries developed B2B EMPs. The in-depth case studies show the types of capabilities that each EMP's executives focused on developing as their EMP passed through several life cycle stages: inception, adaptation, and outcome (Ganesh and Madanmohan, 2004). The analysis identifies the types of capabilities that differentiate B2B EMPs that succeed from those that fail.

3. Methodology

This research employed exploratory case study approaches (Eisenhardt, 1989; Yin, 1994) to answer the two research questions. Following theoretical sampling, two cases were selected based on their similarities and differences. The selected EMPs are referred to as Freight Exchange (“FreightX”) and Convenience Store Exchange (“cStoreX”).

Both EMPs exemplified characteristics representative of other B2B EMPs. These included: (1) executives who envisioned transforming their respective industries by implementing an Internet hub to facilitate frequent, market-based comparisons

using Internet technology (Dai and Kauffman, 2006), (2) the formation period (Day and Fein, 2003), and (3) the ownership structure (White et al., 2007). Second, executives from both EMPs depended on participant contributions to fund the development of their respective EMP and to provide industry expertise to evolve the offerings. Third, the executives from both EMPs provided the researcher access through three stages of development, beginning with inception. These executives experienced differing levels of success, as measured by the relevant industry's willingness to contribute, the value each EMP represented for participants, and each EMP's outcome. The executives who created FreightX generated and sustained substantial participant contributions, and developed FreightX into an EMP that offers members value. FreightX still exists today and mirrors other EMPs worldwide (Dai and Kauffman, 2006) in its structure and strategies. FreightX's offerings include hierarchical trading mechanisms such as private marketplaces, unique prices, long-term contracts, and consulting. The fate of cStoreX was quite different. After years of struggling to generate participant contributions, cStoreX's executives eventually abandoned their dreams of creating an EMP for the convenience store industry. Table 1 lists the cases.

This research involved collecting qualitative and quantitative data using interviews, observations, system demonstrations, and document reviews (Yin, 1994). Table 2 details the interviews conducted for the EMPs, and shows that each case had an embedded research design. Approximately 90% of the interviews were conducted on-site. Interviews ranged from 30 min to 6 h, with 90 min being the most common length. Each completed field visit led to a detailed typed transcript, resulting in about 200 single-spaced transcript pages in addition to 90 pages of external documents (e.g., business plans, sales presentations, and system documentation). The data were analyzed using within-case analyses and cross-case comparison (Eisenhardt, 1989). The within-case analyses resulted in chronologies that described the types of capabilities that the individual EMP's executives focused on during each development phase. Further analysis of these storylines focused on the capabilities that differentiate the EMP that survived from the EMP that failed.

4. Within-case analysis

The two case studies focus on the three development stages through which the EMPs progressed (i.e., inception, adaptation, and outcome) as executives strived to develop B2B EMPs with a critical mass of active members. This within-case analysis identifies the types of capabilities (i.e., inside-out, outside-in, and spanning) that the executives created within their organizations, and shares each EMP's outcome.

4.1. FreightX

4.1.1. Inception

To address under-utilized freight capacity, a former logistics manager started FreightX in 1994 as a public EMP for the highly fragmented US transportation industry. Upon receiving venture capitalist funding, FreightX's executives began working towards their vision of using the Internet to connect shippers and carriers in an online transportation EMP. Freight capacity is an ideal product for electronic trading because it is undifferentiated and perfectly specifiable in terms of price, origin, destination, size, and weight. Fig. 1 depicts FreightX's executives' vision for product and information flow.

In this transportation industry, trading partners conducted transactions in two ways. High-volume shippers with recurring loads developed long-term contracts and embedded relationships with carriers. Low-volume shippers, or shippers with non-recurring loads, searched the market for carriers to haul their loads on an *ad hoc* basis. In these *ad hoc* situations, shippers usually encountered difficulties finding available carriers with the necessary capacity. In turn, carriers' excess capacity was too dynamic to market effectively. This disparity between freight needs and availability created extreme price fluctuations that often forced shippers to spend extended periods of time searching for carriers and comparing prices.

Initially, FreightX's executives tried to remedy this situation by replacing the manual, person-to-person search and negotiation process with a website that presented shipping needs. Shippers would post their shipping needs and the price they were willing to pay. Carriers would then view loads according to geographic region, and accept loads that met their specifications. Initially, FreightX's executives did not charge membership fees in exchange for website access. Rather, FreightX's profit was equal to the difference between the price the shipper was willing to pay and the price the carrier accepted.

FreightX's executives began developing inside-out capabilities when the idea for an EMP formed in 1994. After 5 years, the executives had established a functional EMP that addressed two industry problems: unused capacity, and the inefficient

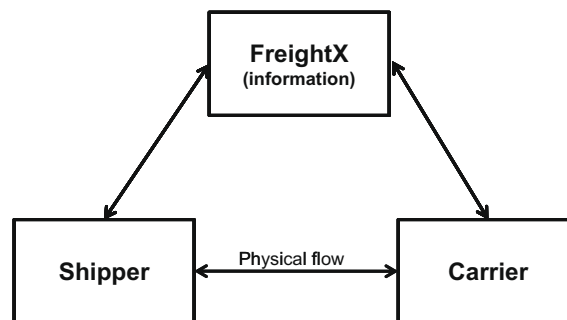
Table 1
EMP cases.

EMP/case name	Industry the EMP served	Participant contribution	Survival
Freight Exchange (FreightX)	Transportation	Over 2500 shippers and carriers joined, executing over 10,000 transactions per day. Contributed industry operation expertise and trading partner relationships	1994-present
Convenience Store Exchange (cStoreX)	Convenience store	Four organizations joined; 3500 BigOil retail convenience stores used daily. The four member organizations contributed \$30 million to fund development, software, industry operation expertise and social capital to solicit members	March 2000–October 31, 2002

Table 2

Interviews conducted for each EMP.

EMP name	Organization ^a	Interviewee job title (No. of interviews) ^b
FreightX	FreightX	President and founder (2 interviews) Regional sales director
	RE Trucking	Bid manager
	Southwestern Trucking	Manager special projects
	Texas Plastics	Distribution manager
	Frosty Brands	Sales representative controller
	Merit Trucking	President (2 interviews)
cStoreX	cStoreX	President and chief operating officer (2 interviews) Chief financial officer cStoreX board member (5 interviews)
	CigMan (Cigarette Manufacturer)	E-commerce representative
	Convenience Store Software Organization	President and founder
	Big Distribution	E-business and international systems director (2 interviews) Marketing vice president (2 interviews) New market development vice president National sales representative
	BigOil	Retail sales manager
	United States Convenience Store Association	Strategic alliances senior vice president
	RetailX EMP	RetailX founder
	Sonoco Retail	Information management director
	SuperSport Retail	Coordinating manager, e-business and emerging technology
	Little Distribution	Director
	Candy Supplier	Sales director
Total number of interviews		33

^a All organization names are pseudonyms.^b Unless otherwise noted, each person represents one interview.**Fig. 1.** FreightX's executives' vision for the transportation industry.

search and negotiation process required for *ad hoc* loads. Unfortunately, few loads were tendered on FreightX because few organizations knew of or had adopted it as their EMP.

4.1.2. Adaptation

In January 2000, FreightX's executives began to develop outside-in capabilities. The executives hired sales managers with transportation industry experience and contacts. FreightX's sales director explained the situation:

"A few years ago, FreightX's executives decided if they were going to go anywhere, they needed to hire people with a knowledge of transportation at the street level. They had technology people with the technology idea but they didn't have anyone that knew how the transportation industry worked at the street level. Hiring people that were already in the transportation industry and getting them to go out and present the idea to the customer was better. I don't come up with a solution and say, 'Here, use it.' I work with the customer. The mentality before they took stock of the need to hire people that knew the marketplace from the ground up, was that B2B was growing so fast, these organizations had to join it to get in the game."

FreightX's sales managers understood that the carrier side of the transportation industry was competitive in that all carriers offer the same product or service. Shippers, on the other hand, represent an array of industries that do not necessarily compete with one another. The sales managers thus opted to market the EMP to their existing shipper contacts. FreightX's sales director explained, "FreightX's sales managers positioned FreightX as a means of becoming involved in electronic commerce (EC) and as a way for IS people to redeem themselves after the year 2000 spending."

After several organizations adopted FreightX, the sales managers realized that FreightX did not fit with existing industry practices. In the transportation industry, nearly 80% of transactions occur through long-term, pre-negotiated contracts. Neither the shippers nor the carriers were willing to give up existing pre-negotiated contracts to take a chance with FreightX. Rather, both sides of the EMP wanted to use FreightX only for *ad hoc* loads and less-than-truckload freight that was not too time sensitive. Unfortunately, these loads did not generate enough transactions for the EMP to tender loads. In fact, FreightX's president indicated that carriers did not pick up 70% of the loads posted on FreightX.

To survive, FreightX's sales managers developed their organization's spanning capabilities by co-opting shippers as co-creators of FreightX. This resulted in several modifications that aligned FreightX's EMP more closely with industry practices. First, FreightX's developers created private trading communities to support bid negotiations, and offered shippers software to manage multiple pre-selected carrier relationships. This software tracked loads down to the stock-keeping unit level, and offered an optimizer program that evaluated and selected freight options based on shippers' pre-specified criteria. After this modification, the participating shippers required that their carriers join FreightX, making the carriers' excess capacity available to the public EMP. Carriers that used FreightX experienced higher profits since FreightX's software evaluated potential loads and optimized their freight capacity.

Second, FreightX's sales managers instituted a quality assurance program through a "three strikes, you're out" rule that applied to both shippers and carriers. Shippers were required to have their loads ready within the time specified; carriers, in turn, were required to be ready to receive the load. In addition, managers verified that carriers were adequately insured, and that shippers were credit worthy. Third, to ensure liquidity, the sales managers moved FreightX beyond merely providing matching and requisitioning services: they began to pre-negotiate contracts among FreightX, shippers, and carriers, and to manage shippers' logistics processes. Rather than using an automated tool, FreightX's sales managers directly negotiated contracts according to which carriers would: (1) haul random loads not tendered on FreightX's open market and/or (2) haul loads for a specific shipper.

4.1.3. Outcome

During this study, business interest in EC dwindled. While many dot-coms were failing, the sales managers were repositioning FreightX as a solution to the next environmental challenge: the economic slowdown caused by the 9/11 terrorist attack. FreightX thus changed its mission from dealing with excess capacity to addressing the freight shortage. Despite many environmental and economic shifts, FreightX survives to this day.

4.2. cStoreX

4.2.1. Inception

Convenience stores are small, geographically dispersed stores that provide sundries to consumers. The convenience store industry's high degree of fragmentation, combined with individual convenience stores' low purchase volume, lead most manufacturers to service convenience stores through industry distributors. The convenience store industry is characterized by poor communication caused partly by retailers' reluctance to adopt technology, and partly by distributors' selective information sharing practices. To address these communication problems, one of the largest retailers in the industry, BigOil, launched cStoreX in March 2000 as a one-stop purchasing hub and online portal. Fig. 2 shows the vision cStoreX's executives had for the product and information flow in the convenience store industry.

Several years before launching cStoreX, BigOil's retail manager had launched an intranet to improve communication between BigOil's 8000+ convenience stores and its corporate office. The same individual, now as cStoreX's president,

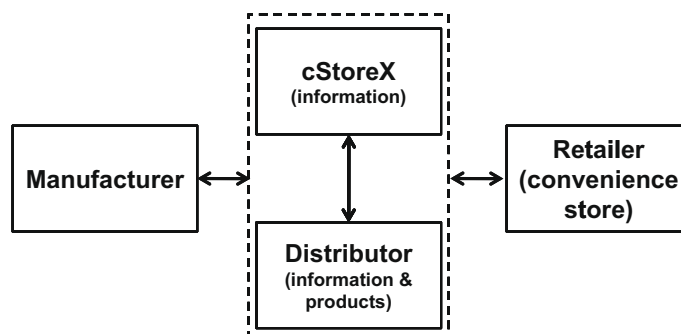


Fig. 2. cStoreX's executives' vision for the convenience store industry.

envisioned building on this success by making BigOil's intranet available to competing retail chains, thus improving communication throughout the industry supply chain. To fulfill this vision, she and the other cStoreX executives focused on inside-out capabilities by hiring cStoreX's workforce and developing its technology. In addition, executives established cStoreX and developed its governance structure by creating an alliance with the industry's leading distributor (Big Distribution). As the leading retailer, BigOil solicited Big Distribution's membership when the distributor was bidding on BigOil's service agreement.

4.2.2. Adaptation

After the inception stage, cStoreX's software engineers continued to develop cStoreX's technology. However, the executives primarily focused on outside-in capabilities by trying to attract a critical mass of retailers and manufacturers to the EMP. As the traditional middleman, Big Distribution had relationships with most industry retailers and manufacturers; cStoreX's president relied on these relationships to attract organizations. Immediately after co-founding cStoreX, Big Distribution invited CigMan, the largest cigarette manufacturer in the United States, to join the cStoreX alliance. CigMan accepted the invitation, and both CigMan and Big Distribution contributed \$10 million in cash and \$10 million in intellectual capital to cStoreX.

With industry leaders on board, cStoreX's executives launched a membership recruitment campaign comprised of media marketing and face-to-face sales calls. cStoreX's executives believed that media reports describing cStoreX's vision, combined with industry leaders' involvement, would stimulate industry adoption. Big Distribution's chief information officer and cStoreX's president met with convenience store industry senior executives to solicit their membership. Unfortunately, although the industry-level relationships were strong, the individuals who attended these sales calls did not personally share these strong relationships. As a result, the sales calls did not focus on how cStoreX could solve these individuals' problems. Instead, they conveyed general information such as cStoreX's vision, technology, revenue plans, and adoption projections. cStoreX refined its revenue plans to include tiered monthly access fees: retailers would pay \$100 monthly for each store; distributors and manufacturers paid based on their sales volume.

Because cStoreX's offerings did not fit with existing industry practices, its executives encountered challenges in their bid to attract members. First, retailers did not want to participate in an EMP formed by their competitor. cStoreX's president tried to address this by legally separating cStoreX from its founder, BigOil. However, retailers remained dissatisfied. Second, cStoreX's president wanted the distributor to require its customers to use the EMP, but Big Distribution refused. Big Distribution competed intensely for retailer contracts, partially by doing business with retailers in whatever way the retailers dictated. In addition, there was no incentive for Big Distribution to push the EMP since distributors benefit from poor information in the convenience store industry and control costs by limiting retailer product choice.

By February 2001, the convenience store industry was losing interest in EC and EMPs. To survive, cStoreX's executives focused on inside-out capabilities as well as outside-in capabilities. cStoreX's president even tried to develop the organization's spanning capabilities. Between February and June 2001, the executives focused on inside-out capabilities as they planned a merger with a competing EMP, and then repositioned cStoreX when the merger collapsed. In the period following the merger's collapse, cStoreX's president focused on spanning capabilities by asking the board members¹ what was required to secure additional funding. Rather than suggesting services that industry leaders would use, the board required that cStoreX further develop its software and that cStoreX's customer, BigOil, rank cStoreX as "superior" on a customer satisfaction survey. When cStoreX met these requirements, the board members each reluctantly donated an additional \$2.5 million to cStoreX, and repositioned cStoreX as a software company that offered private EMPs. The private EMPs allowed each organization to manage its convenience stores and conduct business with its pre-selected trading partners at pre-negotiated prices.

In July 2001, cStoreX's executives focused on outside-in capabilities and began marketing its software. cStoreX's president attended the industry trade show, made face-to-face sales calls with convenience store chains, tried to convince manufacturers to offer their products on the EMP, and urged industry software providers to partner with cStoreX. Most organizations would not buy cStoreX's software because BigOil was the only cStoreX investor using the software. During sales calls, potential customers frequently commented, "Your investors aren't using it, so why should we?" In August 2001, cStoreX tried again to develop spanning capabilities by working with its investors to manage the manual cigarette promotion process. cStoreX's president requested that each investor donate members from several functional areas to collaborate on this software. The investors donated fewer people than cStoreX requested, cautiously shared information, and plainly stated that they would not purchase the software. cStoreX's investors had lost interest.

4.2.3. Outcome

In January 2002, BigOil replaced its cStoreX board member, who represented the cStoreX user group, with an executive who had opposed cStoreX from the start. The BigOil executive made efforts to close or sell cStoreX. cStoreX ceased operations on October 31, 2002. As of the writing of this paper, cStoreX's board member companies have each implemented a proprietary system to address the communication and order processing issues that cStoreX was designed to solve.

¹ cStoreX's board represented the leading organizations in each part of the convenience store industry supply chain. Board members were also cStoreX investors and included representatives from BigOil, Big Distribution, and CigMan.

5. Discussion

This study investigated two B2B EMP development efforts with the aim of understanding the capabilities necessary to develop B2B EMPs that generate and sustain participant contributions. This section summarizes the insights gained from answering the two research questions.

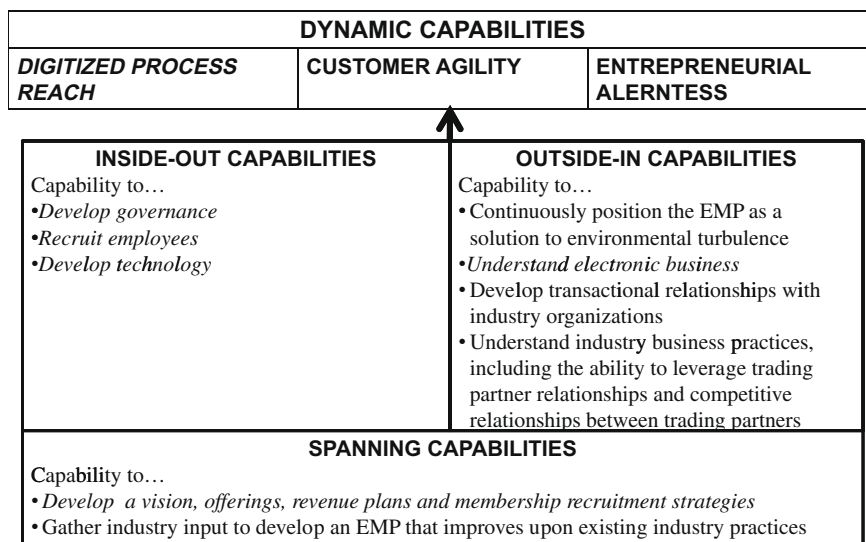
5.1. Differentiating capabilities

The first research question asked, “What types of capabilities differentiate organizations that create B2B EMPs that are able to generate and sustain participant contributions from those that do not?” Fig. 3 identifies the key capabilities that the executives who created FreightX, the successful B2B EMP, developed. The framework indicates that, taken together, the executives’ strategies created dynamic capabilities. These capabilities allowed FreightX’s executives to create an EMP that generated and sustained participant contributions. The text in *italics* indicates capabilities that the executives of cStoreX, the B2B EMP that failed, also developed. While both B2B EMPs’ executives developed inside-out capabilities, FreightX’s executives’ ability to develop outside-in and spanning capabilities differentiated it from cStoreX. The following paragraphs discuss the differentiating capabilities.

5.1.1. Outside-in capabilities

FreightX’s executives developed several interrelated outside-in capabilities that differentiated it from cStoreX. FreightX began generating and sustaining participant contributions when its executives hired sales managers who drew on their relationships and understanding of the industry to solicit organizations for membership. During sales calls, the sales managers characterized FreightX as a solution to problems created by environmental turbulence, such as the need to reduce technology costs after the year 2000 spending, implement Internet technology in response to the dot-com boom, and find freight after the post-9/11 economic slowdown. While cStoreX’s executives initially characterized its offerings as a way to respond to technological change, they were unable to reposition the EMP during the dot-com bust when businesses were more skeptical about the need to invest in new technology.

A second key outside-in capability that differentiated FreightX from cStoreX was the former’s ability to develop transactional relationships with industry organizations. FreightX’s sales managers showed their industry contacts how FreightX could help them meet their performance objectives by reducing shipping costs and increasing efficiency. cStoreX’s executives should have been able to develop transactional relationships with industry organizations since its founders were supply-chain partners engaged in daily, routine transactions. However, Big Distribution’s mixed motivations prevented cStoreX from leveraging these transactional relationships. First, Big Distribution co-founded cStoreX in order to win BigOil’s service contract, not because it wanted to transform the convenience store industry with a B2B EMP. Second, Big Distribution had a



Key:

Italics indicates capabilities that were present in the organizations developing both B2B EMPs

Non-italics indicates capabilities that were substantially more developed in the organization creating FreightX

Fig. 3. Capability development: a contrast between the two B2B EMPs.

long history of doing business with retailers and manufacturers in whatever way these trading partners dictated. Therefore, Big Distribution would not require its trading partners to use cStoreX.

A third outside-in capability that differentiated FreightX from cStoreX was FreightX's sales managers' understanding of industry business practices, including the ability to leverage trading partner relationships and competitive relationships among trading partners. Both EMPs were implemented in fragmented, competitive industries. This rivalry created competitive relationships among trading partners such that organizations were unwilling to share information with an EMP out of fear it would help their competitors. FreightX's sales managers understood transportation industry competition and therefore worked with shippers instead of carriers to develop the EMP. Once the shippers joined FreightX, the carriers joined to secure their shipping contracts. In contrast, cStoreX's executives solicited retailers without recognizing the potentially detrimental effect of the fierce competition among them.

5.1.2. Spanning capabilities

Both EMPs' employees focused on spanning capabilities. However, FreightX's sales managers developed the ability to gather industry input and develop an EMP that improved upon existing industry practices by co-opting transportation managers as co-creators of FreightX. The resulting redesigned EMP incorporated hierarchical mechanisms such as private EMPs with tiered tendering. Despite cStoreX's executives' attempts to develop this capability, they received limited input.

5.1.3. Dynamic capabilities

Fig. 3 suggests that the bundle of inside-out, outside-in, and spanning capabilities created the higher-order dynamic capabilities of digitized process reach, customer agility, and entrepreneurial alertness, which enabled FreightX's executives to generate and sustain participant contributions. Both FreightX's and cStoreX's executives developed digitized process reach in the form of functional EMPs. In contrast to cStoreX's executives, FreightX's sales managers developed a customer agility capability, which allowed them to co-opt customers. FreightX's executives developed an entrepreneurial alertness capability, which allowed their sales managers continuously to explore the marketplace and architect competitive actions. The section that follows explains how FreightX's executives and sales managers developed these capabilities.

5.2. How capabilities are developed

Table 3 highlights the actions FreightX's executives took to develop the dynamic capabilities that differentiated it from cStoreX. The discourse that follows contrasts the actions of the executives who developed the two EMPs and provides evidence that the actions summarized in Table 3 partly explain FreightX's success and cStoreX's failure.

5.2.1. Customer agility

FreightX's executives created customer agility by hiring sales managers and assigning them the task of helping customers use the EMP. Since they had spent their careers working as carrier sales managers, FreightX's sales managers were adept boundary spanners who knew how to balance the competing needs of their organization and their customers. Drawing on their long-term shipper relationships, FreightX's sales managers became familiar with shippers' needs and adapted FreightX to solve their problems. In time, the shippers brought the carriers to the EMP by helping their carriers learn about and implement FreightX. FreightX thus combined the sales managers' past relationships and daily activities to create a customer-centric culture that fostered trust and motivated customer involvement.

cStoreX provides a contrast. Instead of hiring trusted sales managers to help customers use the EMP, cStoreX's president relied on executives from companies that founded cStoreX to promote the cStoreX vision. The executives had spent their careers running internal operations at their organizations and did not have the boundary-spanning experience possessed by FreightX's sales managers. As a result, these executives usually placed the needs of their individual organizations above those of the EMP and its potential customers. This is illustrated in the actions of cStoreX's boundary spanners and in their reluctance to entertain customer requests.

Void of strong personal, inter-organizational relationships, cStoreX's boundary spanners relied on Big Distribution to set up meetings aimed at soliciting industry organizations into the EMP. These meetings, which were usually held with executives in newly created EC positions, focused on general cStoreX corporate information rather than factors unique to each

Table 3
How B2B EMPs develop dynamic capabilities.

Capability	Actions
Customer agility	Hiring operational-level employees with existing industry relationships characterized by trust Assigning operational-level employees to help customers use the EMP
Entrepreneurial alertness	Motivating employees to "do what it takes" to secure business Fostering a "trial-and-error" culture

industry organization. Unfortunately, two dynamics prevented cStoreX from developing customer agility. First, the industry representatives who attended the meetings did not possess the authority to convince buyers and/or sellers in their organization to use the EMP. Second, cStoreX's executives were more focused on leading the industry into the EMP than on creating an EMP for the industry. This is evident in the executives' reluctance to create an open EMP where retailers, distributors, and manufacturers could participate equally. Rather, cStoreX's founders insisted that other industry distributors not participate in the EMP, and created tiered manufacturer privileges. This resulted in a closed EMP, where retailers serviced by competing distributors would not join the EMP and the industry trade association would not endorse it.

5.2.2. Entrepreneurial alertness

These words, spoken by a FreightX sales manager, show how FreightX achieved entrepreneurial alertness:

"We still have a start-up mentality. We do whatever it takes to get a customer to use us. All we want to do is haul freight. In some cases, we don't even make money just because we want the freight."

FreightX's executives clearly espoused a "trial-and-error" culture by empowering its sales managers with the mantra that their job was to move freight. This operating flexibility enabled the sales managers to take a variety of actions, including: (1) negotiating long-term contracts with carriers to haul pre-specified loads and/or freight at pre-specified prices, (2) allowing shippers to use FreightX's software to manage relationships with their existing shippers, and (3) serving as a third-party logistics provider for shippers.

cStoreX provides a contrast. Other than the software engineers, most of cStoreX's employees were loaned to cStoreX by its founding organizations, which were also paying these employees' salary. As a result, these employees were primarily motivated to protect their organizations' competitive positions. This is evident in cStoreX's president's explanation that cStoreX's repositioning strategy failed because its investors held back their input and themselves planned to develop a system similar to cStoreX's EMP:

"But of course, Bill [CIO of Big Distribution and cStoreX board member] wouldn't want to buy it [the cStoreX software]. He was the IT guy; that is what he does is make it. The idea behind this was to deploy the software to the companies with training. The idea was that this would help us have a bigger pie by improving the supply chain in the industry. But no one wanted to admit that it would help them. They were all worried about cutting the pie so they got the right size piece. This was called Smart Promo. While building this, all three companies said, 'This is not a commitment. Until your community is big, we are not going to use it.' It was a catch-22. Until they used it, the community was not going to get big! CigMan changed midstream and stopped collaborating. I think they [CigMan] are going to build a Promo.Cm.Com and require everyone to use their own marketplace."

6. Implications

Tracing the evolution of two B2B EMPs, this study extends the current understanding of EMPs by identifying capabilities that lead to EMP success, and shows how EMP executives develop dynamic capabilities. Customer agility and entrepreneurial alertness play a critical role in EMP sustenance, and executives can develop these dynamic capabilities by hiring sales managers with existing industry relationships, assigning these employees the task of helping customers use the EMP, and motivating employees to "do what it takes" to secure business. While many have called for investigations into the capability development process (Oliver, 1997; Melville et al., 2004; Wade and Hulland, 2004), this is one of the first studies to highlight the critical role of non-executive boundary spanners in EMP development.

These insights have implications for research and practice. While the majority of case studies in this area use secondary, published data for their empirical base (Hess and Kemerer, 1994; Lee and Clark, 1996; Granados et al., 2006; Soh et al., 2006), this paper offers an alternative source of guidance for managers. The examples and framework presented here could function as a roadmap for launching IS projects that specifically depend on participation from a critical mass of industry organizations.

This paper also makes several theoretical advancements. First, this research extends work on dynamic capability development (Montealegre, 2002; Sambamurthy et al., 2003; Leidner et al., 2009) by showing how dynamic capabilities are developed in situations where technology value depends on continued participation from many organizations. Existing research has identified organizational capabilities such as long-term perspective and fostering management skills over time (Mata et al., 1995; Montealegre, 2002). This study adds to this body of knowledge by suggesting that pre-existing relationships and trust are also extremely important to creating B2B EMPs.

Second, this research highlights two qualities of effective boundary spanners in EMP contexts: such individuals have inter-organizational relationship skills and are operational-level employees, strengths recognized by the successful EMP's executives. In a similar vein, Levina and Vaast (2005) found that less distinguished people make better boundary spanners. Taken together, these findings may suggest that previous findings indicating that management skills confer a competitive advantage (Mata et al., 1995; Eisenhardt and Schoonhoven, 1996) may not tell the whole truth. Boundary spanning skills may be more relevant in an EMP context.

With the exception of Leidner et al. (2009), most dynamic capabilities research has been conducted at the single organization level of analysis. Thus, the third contribution of this paper is that it adds to the nascent body of knowledge on dynamic capabilities research applied at the inter-organizational level. As an extension of Wade and Hulland's (2004) proposition that

outside-in and spanning capabilities may be more important for organizations in turbulent, complex environments, this research underscores the need for managers to pay particular attention to developing outside-in and spanning capabilities in cooperative contexts. While both EMPs developed inside-out capabilities, the development of spanning and outside-in capabilities differentiated FreightX from cStoreX. Future research on IT capability development at the inter-organizational level could strengthen these insights further.

Finally, this study demonstrates the critical role that the environment and the industry play in EMPs' ability to develop capabilities. In particular, the study supports research on IT fashion by showing that organizations' willingness to contribute to EMPs was somewhat dependent on electronic business fads (Swanson and Ramiller, 2004) and the industry's competitive history. This finding responds to calls to consider the role that the environment and the industry play in capability development (Montealegre, 2002; Wade and Hulland, 2004).

7. Limitations and conclusion

This study's contributions need to be considered in light of its limitations. First, while the analysis provides a reasonable explanation for FreightX's success and cStoreX's failure, alternative explanations cannot be discarded. It is plausible that the complexity of the convenience store industry and/or cStoreX's potential disruption of the industry power structure may have contributed to cStoreX's failure. Second, although contact with key informants is ongoing, most of the case study data were collected several years ago. However, EMPs continue to play a vital role in supply chains and the managerial implications for developing industry collaborative networks remain relevant. Third, the in-depth focus on just two EMPs may raise questions about generalizability. However, the details included in each case, along with the iterations between the cases and the literature, suggests that the findings reflect some general patterns. As such, the two models developed in this study (i.e., Fig. 3 and Table 3) should be testable and empirically valid in other settings where technology success depends on the collaboration and participation of a critical mass of industry organizations. Future research is necessary to add insight into how to develop the capabilities necessary for EMP survival.

Despite these limitations, this study should be of interest to managers developing procurement technology that requires ongoing collaboration and participation from a variety of industry players. This research revealed that spanning and outside-in capabilities might be more difficult to develop than inside-out capabilities. Both EMPs in this study developed inside-out capabilities and digitized process reach. FreightX, however, was successful because it also developed customer agility and entrepreneurial alertness. This research develops a framework that reflects these findings. Furthermore, the research proposes ways in which customer agility and entrepreneurial alertness can be developed to complement digitized process reach and create EMPs that generate and sustain participant contributions. Hopefully, these insights will spark interest among IS scholars to examine capability development more closely in situations where technology can serve as a public good for an entire industry.

Acknowledgements

I am thankful for the guidance and support provided by Dorothy Leidner, Claudia Loebbecke, the two reviewers, Baylor University, Texas A&M University and my industry contacts.

References

- Allen, D.K., Colligan, D., Finnie, A., Kern, T., 2000. Trust, power and interorganizational information systems: the case of the electronic trading community translease. *Information Systems Journal* 20 (10), 21–40.
- Bakos, J.Y., 1997. Reducing buyer search costs: implications for electronic marketplaces. *Management Science* 43 (12), 1676–1692.
- Barrett, M., Walsham, G., 1999. Electronic trading and work transformation in the London insurance market. *Information Systems Research* 10 (1), 1–22.
- Choudhury, V., Hartzel, K.S., Konsynski, B.R., 1998. Uses and consequences of electronic markets: an empirical investigation in the aircraft parts industry. *MIS Quarterly* 22 (4), 471–507.
- Dai, Q., Kauffman, R.J., 2006. To be or not to B2B: evaluating managerial choices for e-procurement channel adoption. *Information Technology and Management* 7 (2), 109–130.
- Day, G., 1994. The capabilities of market-driven organizations. *Journal of Marketing* 58 (4), 37–52.
- Day, G.S., Fein, A.J., 2003. Shakeouts in digital markets: lessons from B2B exchanges. *California Management Review* 45 (2), 131–150.
- Eisenhardt, K., Martin, J., 2000. Dynamic capabilities: what are they? *Strategic Management Journal* 21 (10/11), 1105–1121.
- Eisenhardt, K.M., 1989. Building theory from case study research. *Academy of Management Review* 14 (4), 532–550.
- Eisenhardt, K.M., Schoonhoven, C.B., 1996. Resource-based view of strategic alliance formation: strategy and social effects of entrepreneurial firms. *Organization Science* 7 (2), 136–150.
- Ganesh, J., Madanmohan, T.R., 2004. Adaptive strategies of firms in high velocity environments: the case of B2B electronic marketplaces. *Journal of Global Information Management* 12 (1), 41–59.
- Granados, N.F., Gupta, A., Kauffman, R.J., 2006. The impact of IT on market information and transparency: a unified theoretical framework. *Journal of the Association for Information Systems* 7 (3), 148–178.
- Granados, N.F., Gupta, A., Kauffman, R.J., 2007. IT-enabled transparent electronic markets: the case of the air travel industry. *Information Systems and E-Business Management* 5 (1), 65–91.
- Grewal, R., Corner, J.M., Mehta, R., 2001. An investigation into the antecedents of organizational participation in business-to-business electronic markets. *Journal of Marketing* 65 (July), 17–33.
- Hess, C.M., Kemerer, C.F., 1994. Computerized loan origination systems: an industry case study of the electronic markets hypothesis. *MIS Quarterly* 18 (3), 251–275.
- Howard, M., Vidgen, R., Powell, P., 2006. Automotive e-hubs: exploring motivations and barriers to collaboration and interaction. *Journal of Strategic Information Systems* 15 (1), 51–75.

- Kambil, A., van Heck, E., 1998. Reengineering the Dutch flower auctions: a framework for analyzing exchange organizations. *Information Systems Research* 9 (1), 1–19.
- Lee, H.G., Clark, T.H., 1996. Market process reengineering through electronic market systems: opportunities and challenges. *Journal of Management Information Systems* 13 (3), 113–136.
- Leidner, D., Gary, P., Shan, P., 2009. The role of IT in crisis response: lessons from the SARS and Asian Tsunami disasters. *Journal of Strategic Information Systems* 18 (2), 80–99.
- Levina, N., Vaast, E., 2005. The emergence of boundary spanning competence in practice: implications for implementation and use of information systems. *MIS Quarterly* 29 (2), 335–363.
- Malone, T.W., Yates, J., Benjamin, R.I., 1987. Electronic markets and electronic hierarchies. *Communications of the ACM* 30 (6), 484–497.
- Mata, F.J., Fuerst, W.L., Barney, J.B., 1995. Information technology and sustained competitive advantage: a resource-based analysis. *MIS Quarterly* 19 (4), 487–505.
- Matook, S., Vessey, I., 2008. Types of business-to-business e-marketplaces: the role of a theory-based, domain specific model. *Journal of Electronic Commerce Research* 9 (4), 260–279.
- Melville, N., Kraemer, K., Gurbaxani, V., 2004. Review – information technology and organizational performance: an integrative model of IT business value. *MIS Quarterly* 28 (2), 283–322.
- Mithas, S., Jones, J.L., Mitchell, W., 2008. Buyer intention to use internet-enabled reverse auctions: the role of asset specificity, product specialization, and non contractibility. *MIS Quarterly* 32 (4), 705–724.
- Montealegre, R., 2002. A process model of capability development: lessons from the electronic commerce strategy at Bolse de Valores de Guayaquil. *Organization Science* 13 (5), 514–531.
- Oliver, C., 1997. Sustainable competitive advantage: combining institutional and resource-based views. *Strategic Management Journal* 18 (9), 697–713.
- Premkumar, G., Richardson, V.J., Zmud, R.W., 2004. Sustaining competitive advantage through a value net: the case of enterprise rent-a-car. *MIS Quarterly Executive* 3 (4), 189–199.
- Rangan, K., 1999. Freemarkets online. Harvard Business School Case # 9-598-109.
- Sambamurthy, V., Bharadwaj, A., Grover, V., 2003. Shaping agility through digital options: reconceptualizing the role of information technology in digital firms. *MIS Quarterly* 27 (2), 237–263.
- Soh, C., Markus, M.L., Goh, K.H., 2006. Electronic marketplaces and price transparency: strategy, information technology and success. *MIS Quarterly* 30 (3), 705–724.
- Swanson, E.B., Ramiller, N.C., 2004. Innovating mindfully with information technology. *MIS Quarterly* 28 (4), 553–583.
- Teece, D.J., 1992. Competition, cooperation, and innovation: organizational arrangements for regimes of rapid technological progress. *Journal of Economic Behavior and Organizations* 18 (1), 1–25.
- Teece, D.J., Pisano, G., Shuen, A., 1997. Dynamic capabilities and strategic management. *Strategic Management Journal* 18 (7), 509–533.
- Wade, M., Hulland, J., 2004. Review – the resource-based view and information systems research: review, extension, and suggestions for future research. *MIS Quarterly* 28 (1), 107–142.
- White, A., Daniel, E., Ward, J., Wilson, H., 2007. The adoption of consortium B2B e-marketplaces: an exploratory study. *Journal of Strategic Information Systems* 16 (1), 71–103.
- Yin, R.K., 1994. *Case Study Research: Design and Methods*. Sage Publications, Thousand Oaks, CA.